Adv.

UNSUPERVISED LEARNING →

PCA →

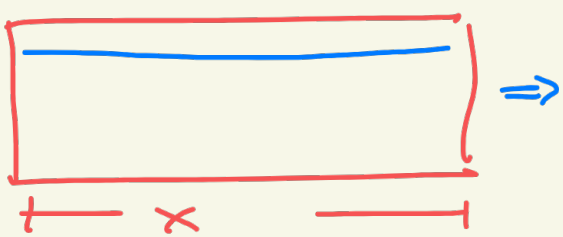
Supervised ⇒

- Dim^n Reduc^n
- Clustering
- Anomaly detection

→ PCA ⇒

UNLABELLED DATA

Ex



Agenda → Problem statement

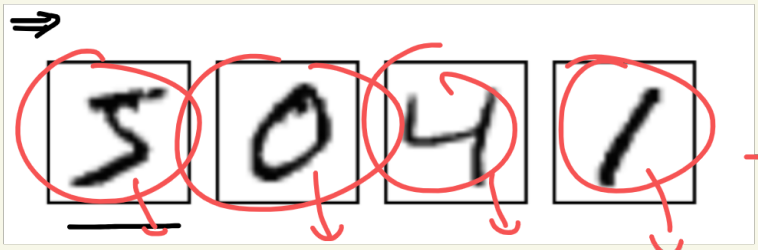
→ Dim^n Reduc^n

→ PCA intuition

→ MATHS →

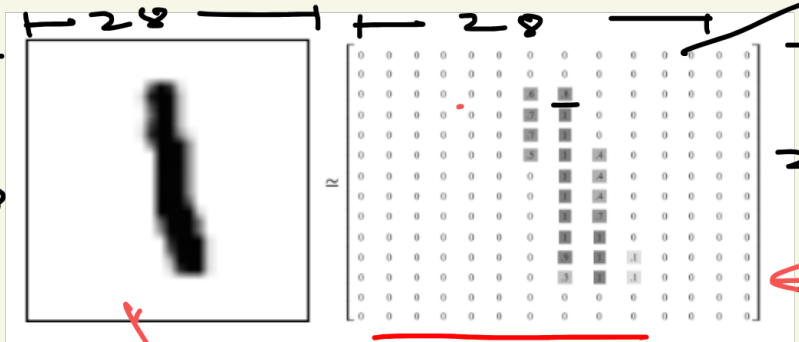
→ Eigenvalues & eigen vectors

Ps → MNIST Digit DATA



0-9

5 0 4 1



Pixel

flatten

28×28

matrix

28 R

28 C

1

(0-255) → [0-1]

dark

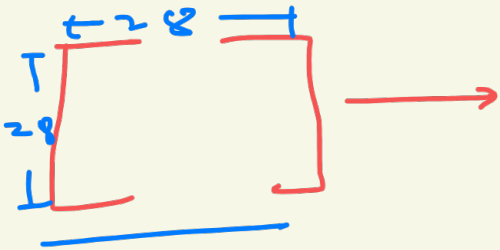
Bright

Dark

Bright

Normalization

min max Scaling



.flatten()

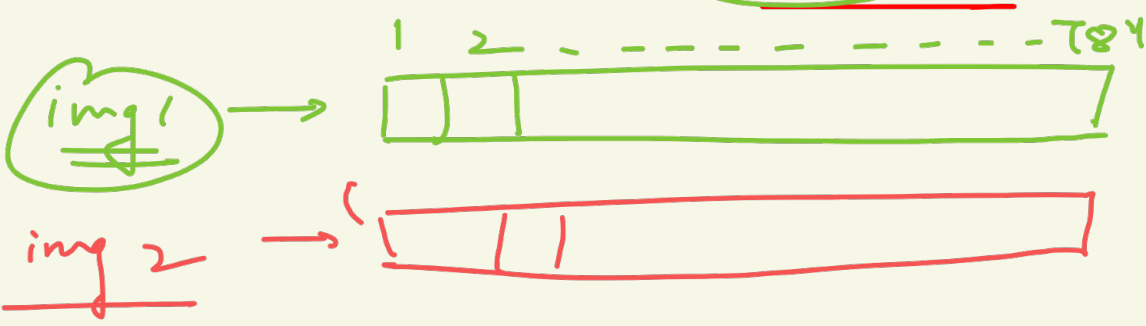
vector → [ ]

1D array

28 × 28 → flatten

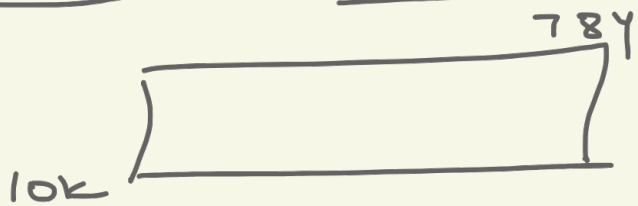
[vector]

784 no's



img → matrix → flatten

10 k imgs → size of data



(10000, 784)

2-d

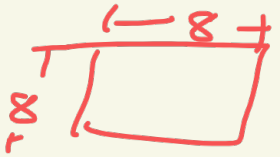
8

no. of img

no. of pixels

SKLEARN → load_digits

8×8



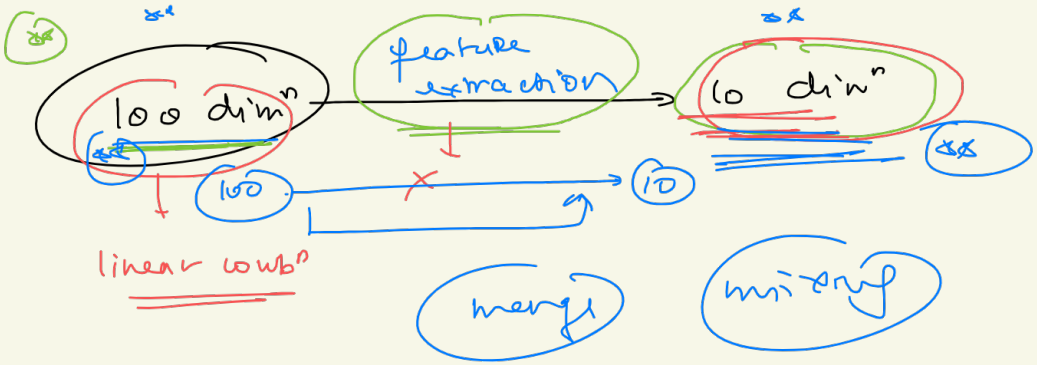
* Dim^n Reduction → Reducing the dim^n

features

i/p data

X Target

column

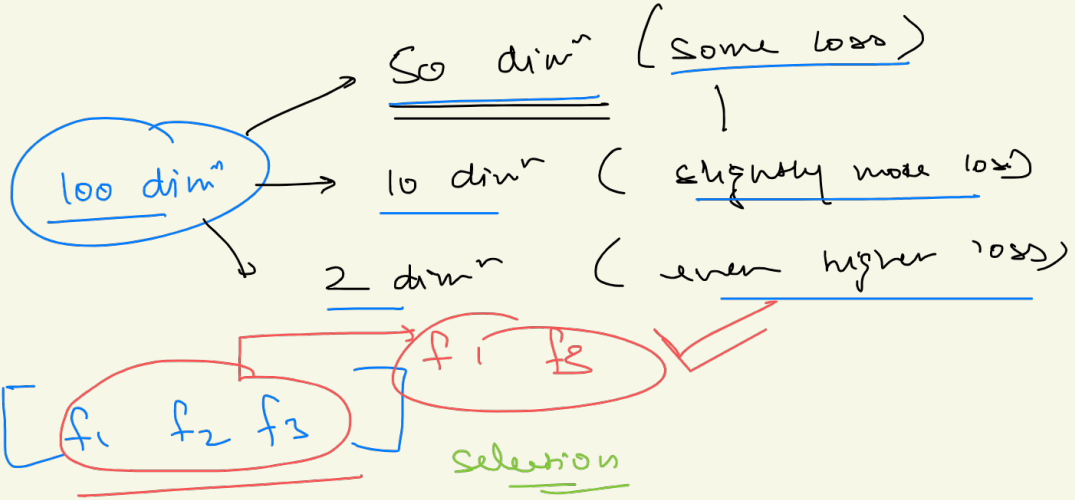


merge

mixing

loss of info ? → loss

→ lossy compression *



f_1 f_2 f_3

Extraction

$f_1 + f_2$ ✓

$f_1 + f_3$ ✓

* Why dim^n Reduction ?

(1) Curse of Dimensionality

data → no. of DP

DATA → grow

Training → Slower

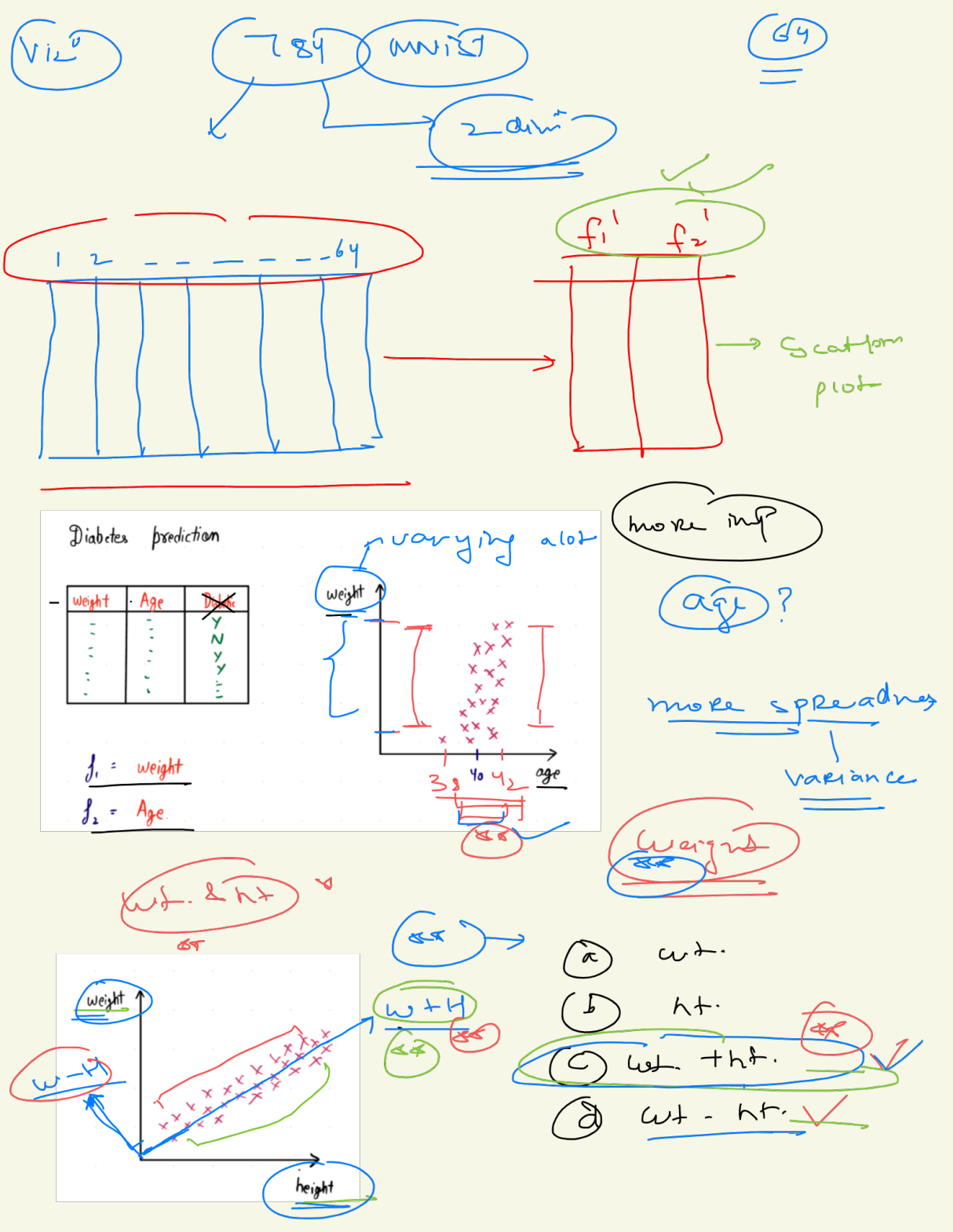
Dim^n Reduc^n

Viz^n

10k features → 500 features

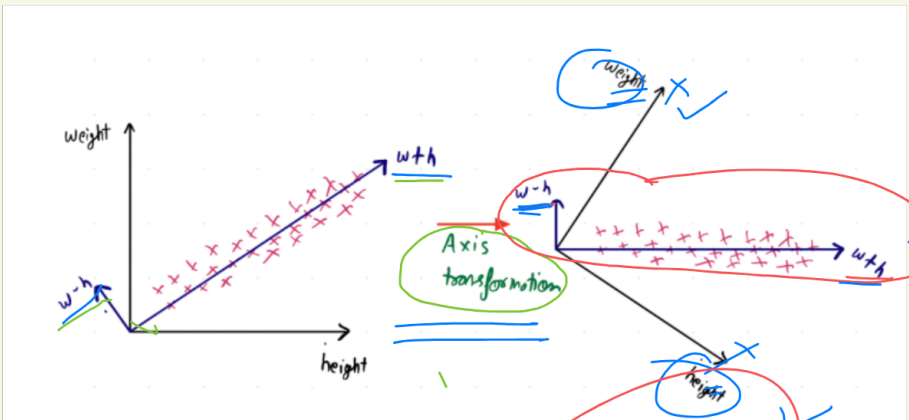
* 1% of loss of info

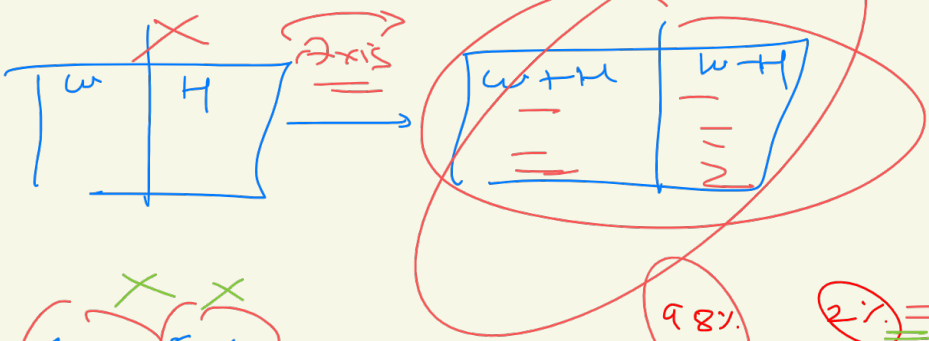
win win ⇒ Training → faster

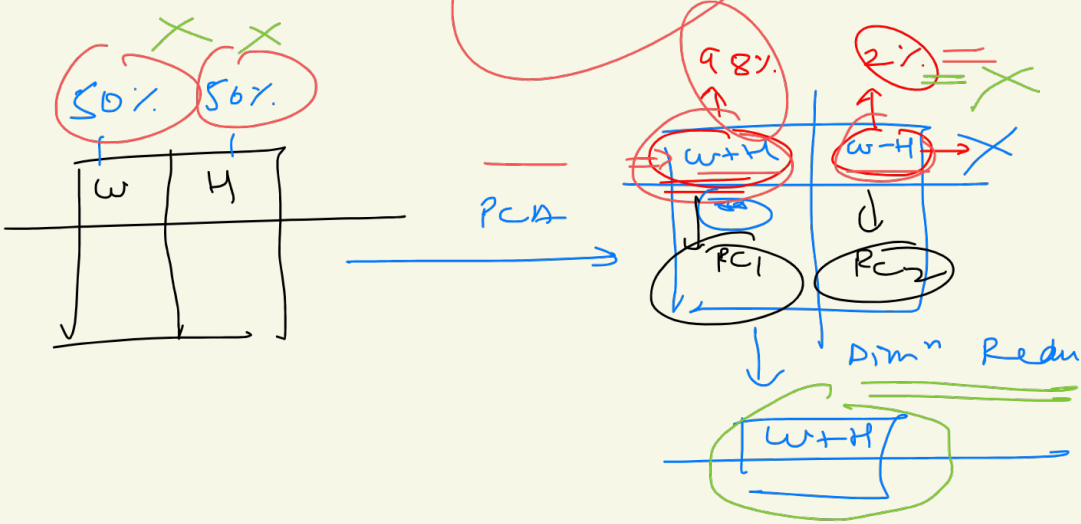

Viz
784
MNIST
2 dim
64
1 2 64
f_1' f_2'
Scatter plot
Diabetes prediction
weight Age Diabetic
Y N Y Y
f_1 = weight
f_2 = Age
varying alot
weight
30 40 42 age
more imp
age ?
more spreadness
variance
Weight
wt. & ht
weight
w+h
w-h
height
a wt.
b ht.
c wt. + ht.
d wt - ht.

PCA → Axis Transformation

1 dimⁿ → Height
2nd → Wt.
→ Transformed axis's



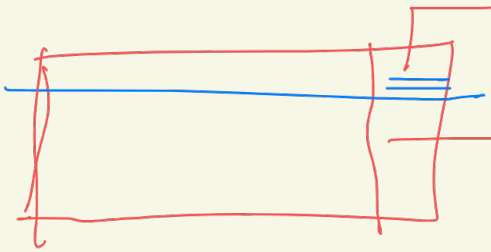




linear combⁿ

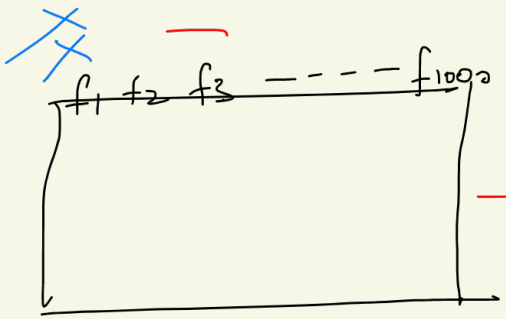
Wt. Ht.

Wt. Ht

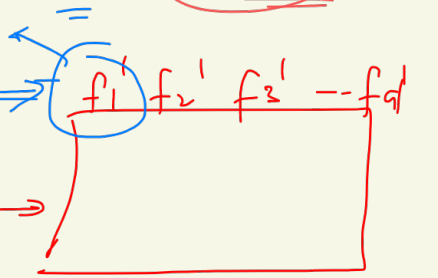


VARIANCE

Target ✗ PCA ✗ label



PCA



new features

$d = 1000$

$d' = 10$

10

Higher variance = more info

PCA

umap

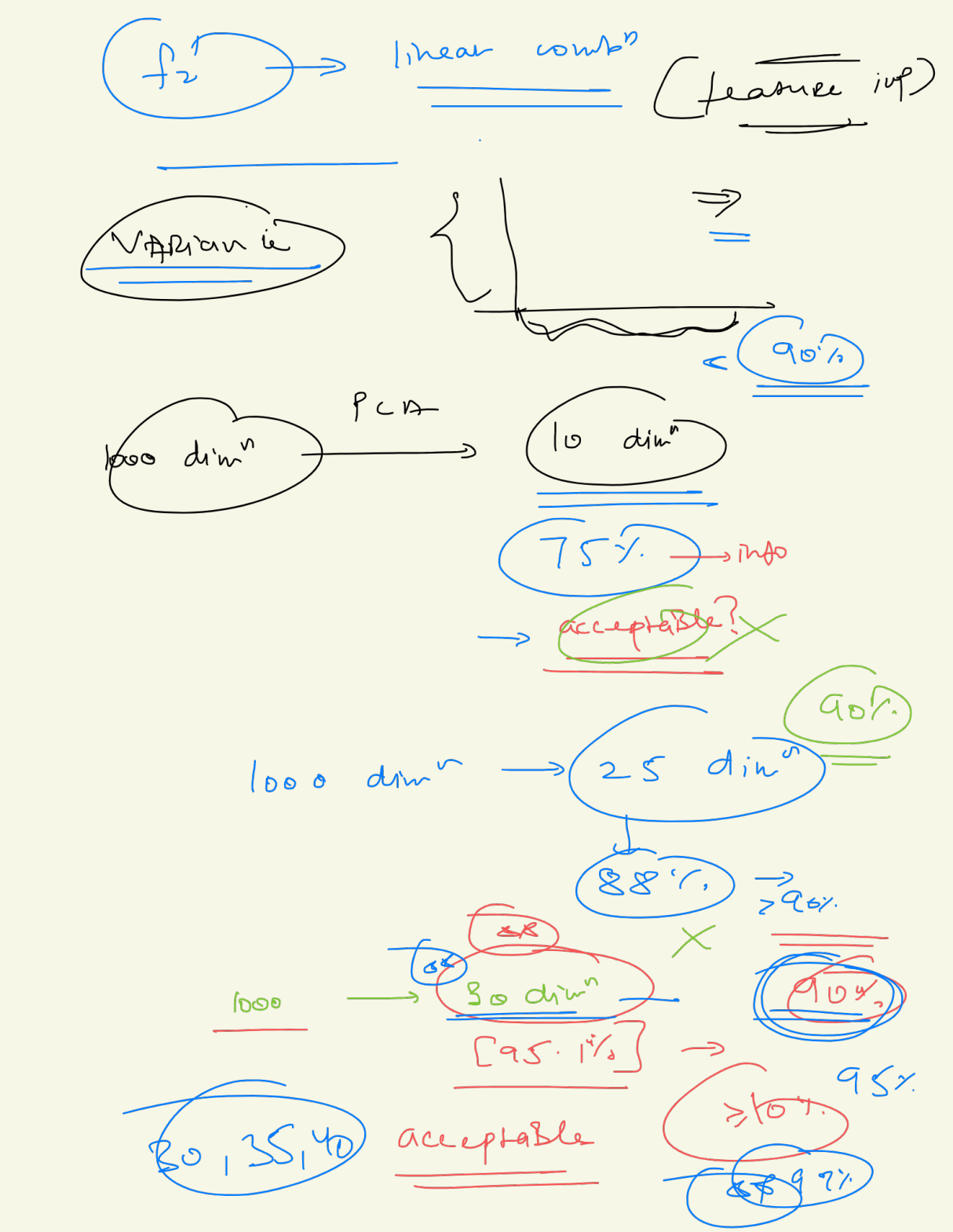
interpretability

tool

training

PCA → linear alg

linear combⁿ


f2
linear combn
(feature imp)
Variance
90%
PCA
1000 dimn
10 dimn
75%
info
acceptable?
90%
1000 dimn
25 dimn
88%
>95%
90%
1000
30 dimn
[95.1%]
95%
>10%
97%
80, 35, 40
acceptable

gaining

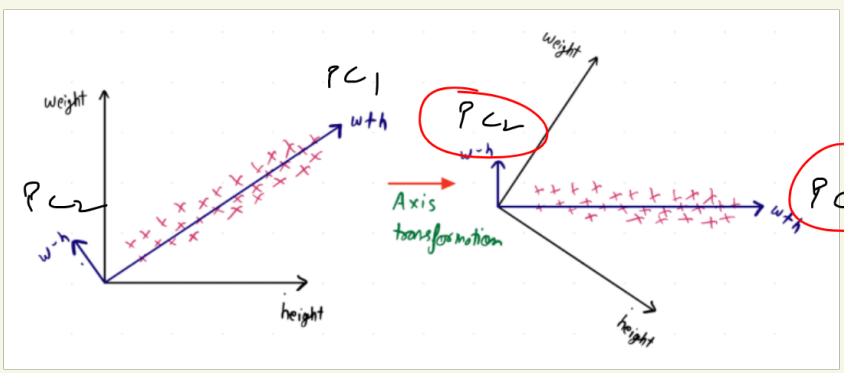
Training → faster

dim ↓ → 112

Break → 10:52

Why PCA → new features !

PCs → principal components



d = 1000

100%.

PCA

1000 × 1000

1000 PCs

n = 10

100%.

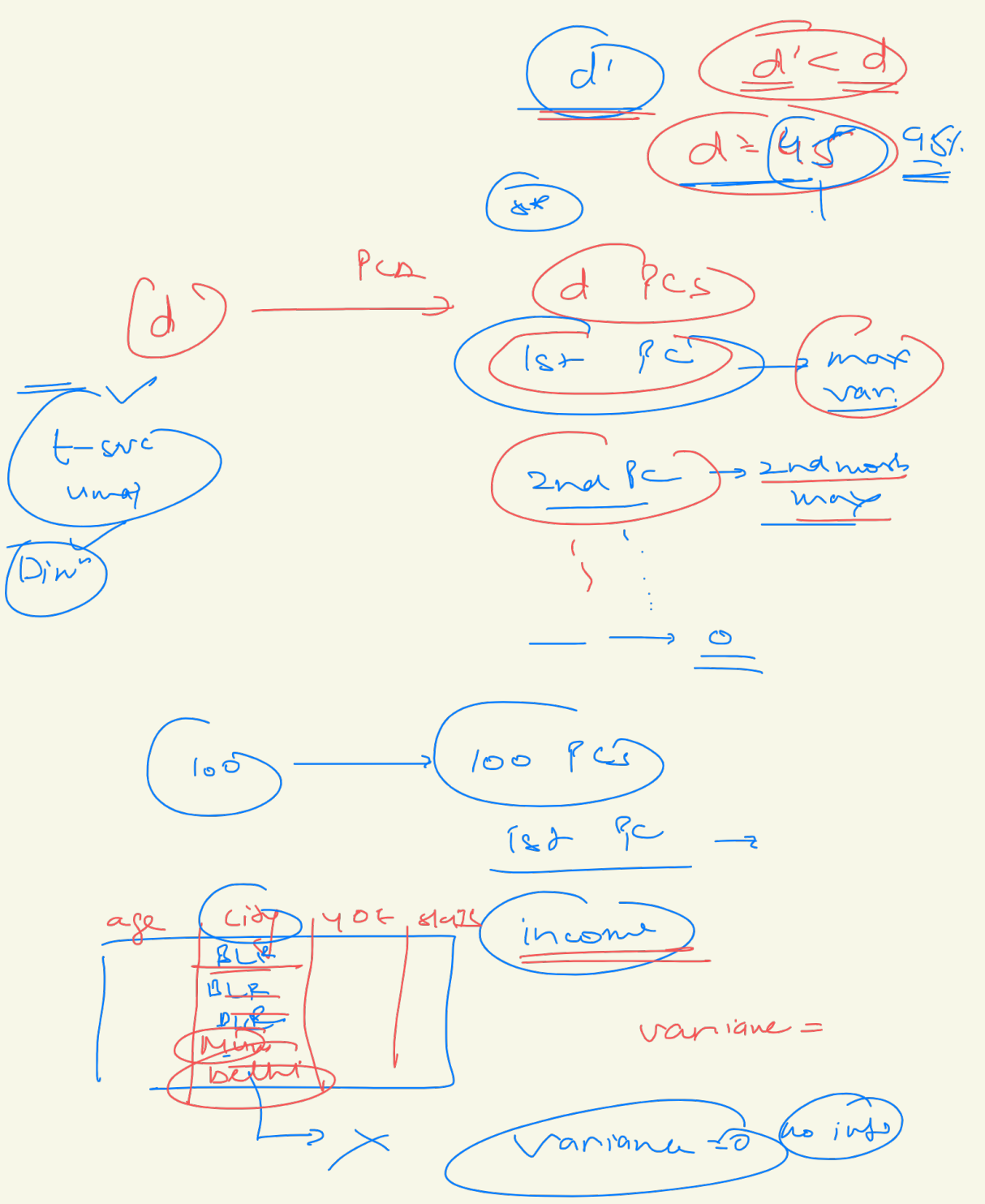
PCA

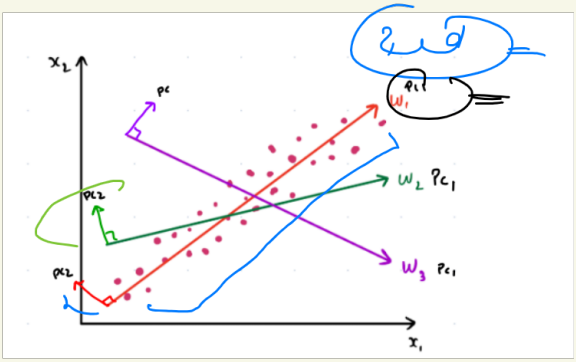
1000 d

d PCs

100% variance

1000


d'
d' < d
d ≥ 95%
d*
PCA
d
d PCs
1st PC
max var.
2nd PC
2nd max var
t-src
Dim
0
100
100 PCs
1st PC
age
city
YOE
income
variance =
variance = 0
no info



Best?

PC

60%

40%

25%

f_2

f_1

pc_1

pc_2

max. variance

maximized

obj

$\theta = 0$

cos θ

x_1

θ

f_1

$$\text{Proj}_{f_1}(x_1) = \frac{x_1 \cdot f_1}{\|f_1\|}$$

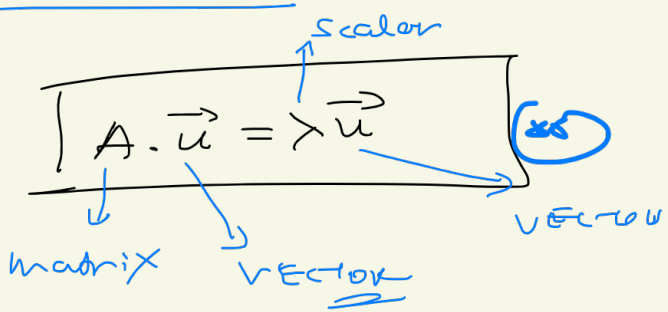
PCs

OBJ → PCs

Eigen values & Eigen vectors ⟹

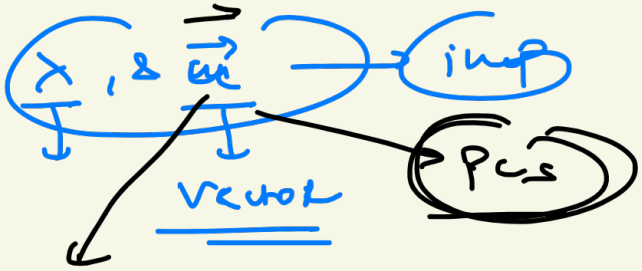
→ Pre Requisite for PCA

for matrix A, there exists one or more than one, vector ($\vec{u}$) such that



⟹ λ = eigen value of A

$\vec{u}$ = eigen vector associated with λ *



$\lambda \rightarrow$ importance of PC

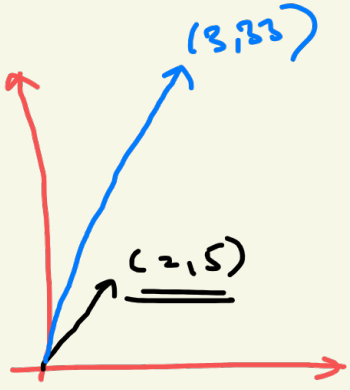
variance

$$A = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \qquad \vec{u} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$A\vec{u} = \lambda\vec{u}$$

$$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 33 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$



λ — eigenvalue

$\begin{bmatrix} 2 \\ 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 8 \\ 33 \end{bmatrix}$

NO possible way

$\vec{u} \times A \Rightarrow$

$= x$

$\vec{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \longrightarrow A = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$

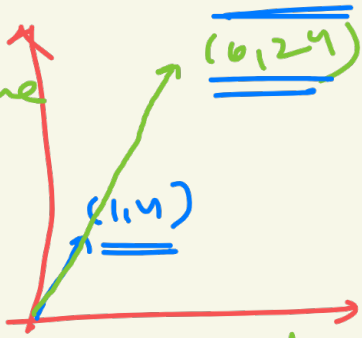
$A \cdot \vec{u} = x\vec{u}$

$\begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$\begin{bmatrix} 6 \\ 24 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$\lambda = 6$

$\lambda = 6$ → eigen value



$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ → eigen vector

Simply Scaling

Matrix multiplication

Vector with matrix →

→ got scaled → eigen vector

$\lambda v = Av$ → eigen value

⇒ $\vec{u}$

$A \begin{bmatrix} \quad \end{bmatrix} \longrightarrow \vec{u}$

$$A = \begin{bmatrix} -5 & 2 \\ -1 & 4 \end{bmatrix}$$

eigen values & eigen vectors

---

(2) $\det(A - \lambda I) = 0$

numpy

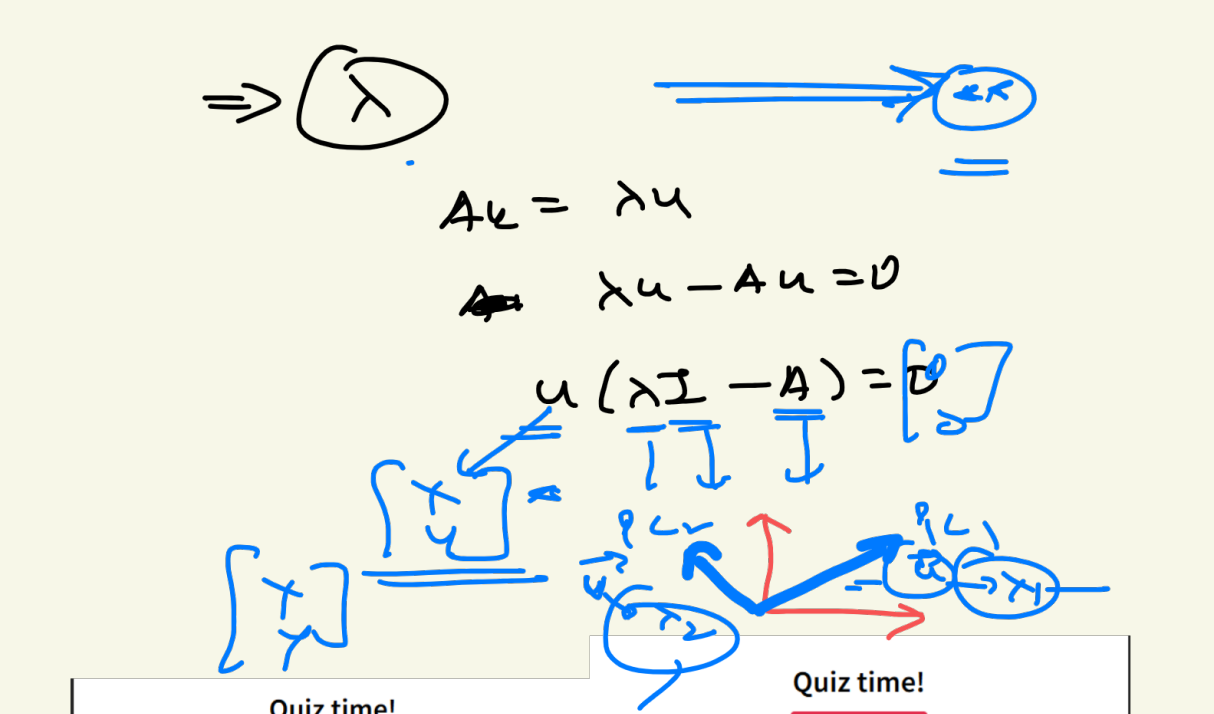
$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\det\left(\begin{bmatrix} -5 & 2 \\ -1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\det\left(\begin{bmatrix} -5-\lambda & 2 \\ -1 & 4-\lambda \end{bmatrix} \right) = 0$$

$(-5-\lambda)(4-\lambda) + 14 = 0$

$\lambda =$



$Au = \lambda u$

$\lambda u - Au = 0$

$u(\lambda I - A) = 0$

Quiz time!

Time Left: 8s

How does PCA ensure that the most information is retained when reducing dimensionality?

23 users have participated

A	By discarding components with the highest variance.	0%
B	By selecting components that are orthogonal to each other.	4%
C	...of	96%
D		0%

Quiz time!

Time Left: 29s

Which statement best describes the property of eigenvectors in the context of PCA?

3 users have participated

A	Eigenvectors change both the magnitude and direction of the original vectors.	0%
B	Eigenvectors change the direction but not the magnitude of the original vectors.	33%
C ✓	Eigenvectors change the magnitude but not the direction of the original vectors.	67%
D	Eigenvectors are unrelated to the direction or magnitude of the original vectors.	0%

Quiz time!

Quiz Ended!

In the context of PCA, what do the eigenvalues represent?

23 users have participated

A	The directions of maximum variance in the data.	13%
B ✓	The amount of variance explained by each principal component.	83%
C	The mean of each variable in the dataset.	4%
D	The sum of squared deviations from the mean.	0%

$\lambda = \max$

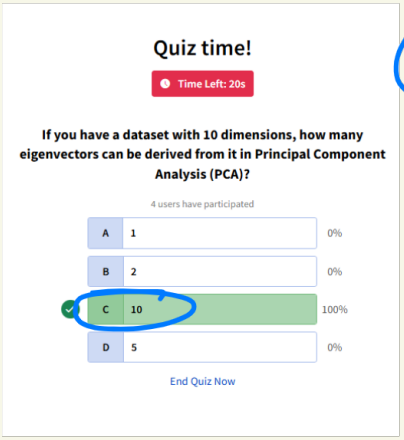
Which eigenvector will preserve the maximum variance in Principal Component Analysis (PCA)?

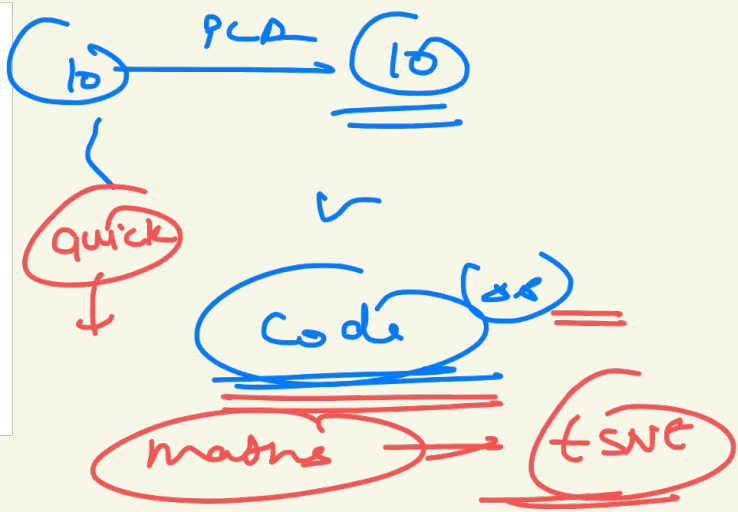
4 users have participated

A	The eigenvector with the smallest eigenvalue	0%
B ✓	The eigenvector with the largest eigenvalue	25%
C	The eigenvector that is perpendicular to all other eigenvectors	0%
D	The eigenvector that is closest to the mean of the data	0%

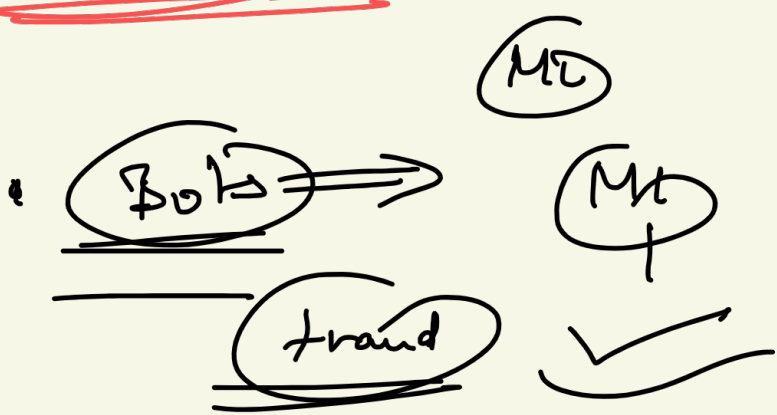
End Quiz Now

λ = SSR


Quiz time!
Time Left: 20s
If you have a dataset with 10 dimensions, how many eigenvectors can be derived from it in Principal Component Analysis (PCA)?
4 users have participated
A 1 0%
B 2 0%
C 10 100%
D 5 0%
End Quiz Now


10
PCA
10
quick
Code
maths
tSNE


Doubt


ML
Bots
ML
train